

CSC 309

Data Structure and Algorithm

Examination: Semester Examination	Semester: Second Semester
Year: 2024	Duration: 2 hours
Total Questions: 5	

Instructions: Answer ALL questions. All questions carry equal marks unless otherwise stated.

Question 1

- (a) Define data structure and algorithm. Explain the relationship between them. (6 marks)
- (b) Analyze the time complexity of the following code snippets using Big-O notation:

```
for(i=0; i for(j=0; j sum = sum + arr[i][j];  
}
```


(5 marks)
- (c) Differentiate between time complexity and space complexity. (4 marks)

Question 2

- (a) Explain the concept of a linked list. Differentiate between singly linked list, doubly linked list, and circular linked list. (8 marks)
- (b) Write an algorithm to insert a node at the beginning of a singly linked list. (7 marks)

Question 3

- (a) What is a stack? Explain the PUSH and POP operations with diagrams. (7 marks)
- (b) Convert the following infix expression to postfix using a stack: $(A + B) * (C - D) / E$. Show all steps. (8 marks)

— End of Examination —